

Topic 6

Frequency Response of FIR Filters

- Sinusoidal response of FIR systems
- Magnitude and phase response
- Delay, first difference, cascades, moving average
- Filtering sampled continuous-time signals

$$H(e^{j\omega})$$

Key idea. Frequency response tells how an FIR filter changes each sinusoidal component.

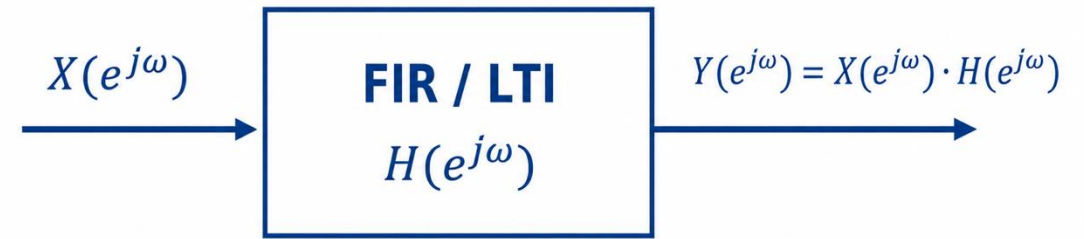
Sinusoidal response of FIR systems

Complex exponentials are eigenfunctions of LTI filters

$$y[n] = \sum_{k=0}^M b[k] x[n - k]$$

$$x[n] = X(e^{j\omega}) e^{j\omega n}$$

$$H(e^{j\omega}) = \sum_{k=0}^M b[k] e^{-j\omega k}$$



same frequency, new amplitude and phase

- The frequency does not change.
- The complex gain H changes amplitude and phase.

Key idea. For LTI systems, sinusoids pass through as sinusoids at the same frequency.

From $H(e^{j\omega})$ to real sinusoidal output

Magnitude scales; phase shifts

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$

$$A \cos(\omega n + \varphi) \rightarrow A |H(e^{j\omega})| \cos(\omega n + \varphi + \angle H(e^{j\omega}))$$

- The output amplitude is multiplied by the magnitude response.
- The output phase is shifted by the phase response.
- A zero in H cancels that sinusoidal component.

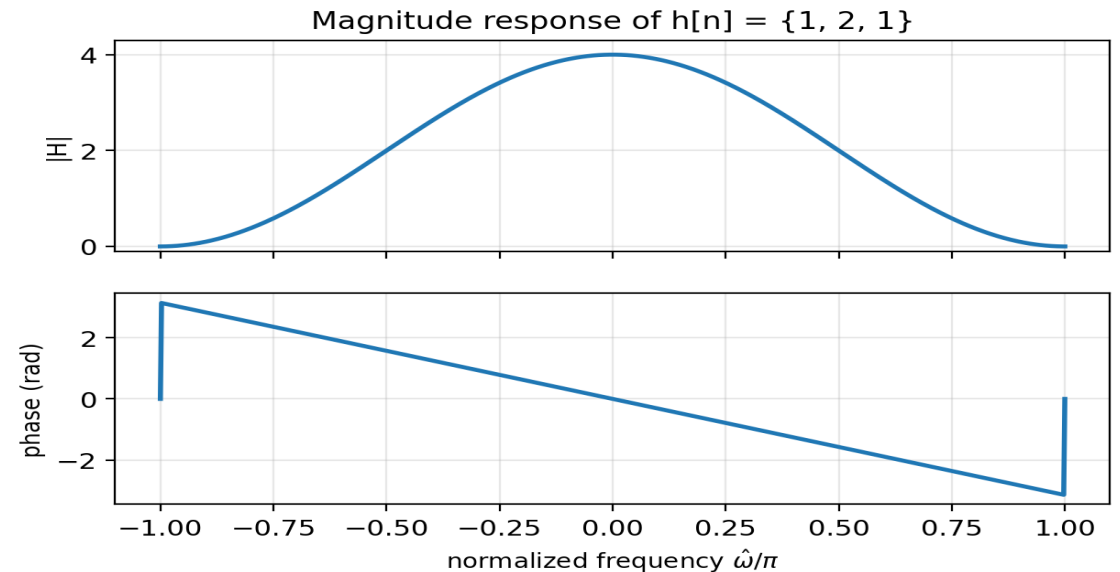
Key idea. The frequency response is a lookup table for every sinusoidal component.

Example 1: sinusoidal response

Find H and the steady-state output

$$y[n] = x[n] + 0.5x[n-1], \quad x[n] = 4\cos(\pi n/2)$$

- 1 $H(e^{j\hat{\omega}}) = 1 + 0.5e^{-j\hat{\omega}}$.
- 2 At $\hat{\omega} = \pi/2$: $H = 1 - 0.5j = 1.118e^{-j0.464}$.
- 3 Output amplitude: $4 \cdot 1.118 = 4.472$.
- 4 Final output: $y[n] = 4.472 \cos(\pi n/2 - 0.464)$.



Key idea. Evaluate H at the input frequency; then scale and shift the cosine.

Superposition and frequency response

A sum of sinusoids is filtered component by component

$$x[n] = \sum_r X_r e^{j\hat{\omega}_r n}$$

$$y[n] = \sum_r X_r H(e^{j\hat{\omega}_r}) e^{j\hat{\omega}_r n}$$

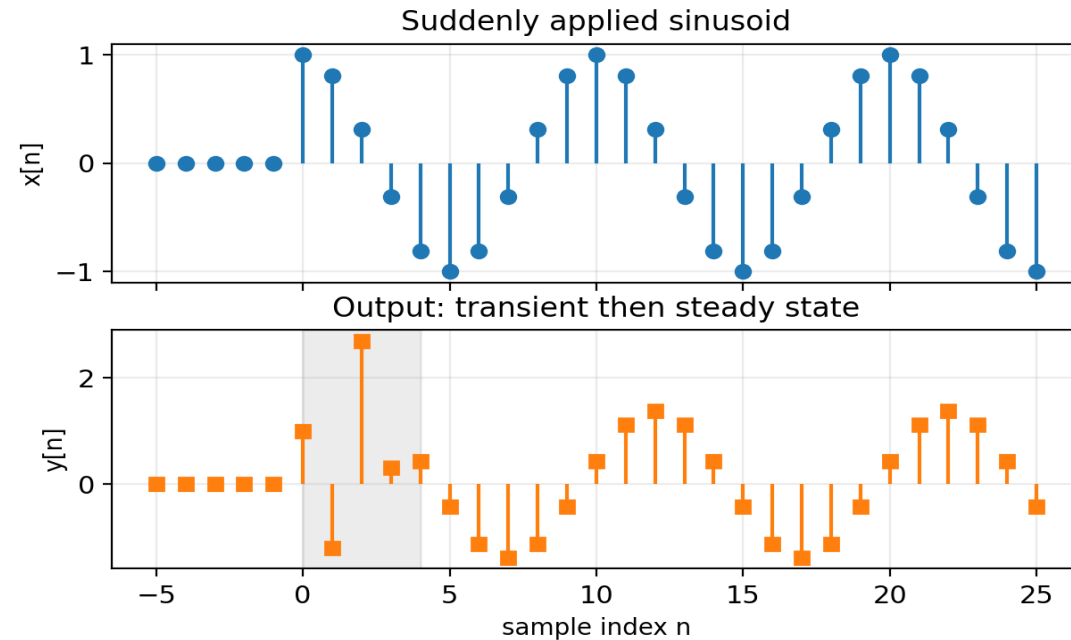
- Linearity lets us filter each component separately.
- No new frequencies are created by an LTI FIR filter.
- Components can be amplified, delayed, attenuated, or cancelled.

Key idea. Frequency response maps an input spectrum into an output spectrum.

Steady-state and transient response

A finite-onset sinusoid is not an eternal eigenfunction

- Before the input begins, the output is zero.
- During the first M samples, the FIR memory is only partly filled.
- After the transient, the sinusoidal steady-state formula applies.
- When the input ends, a second transient occurs.



Key idea. The sinusoidal-response rule describes the steady state, not the onset transient.

Properties of the frequency response

Periodicity and symmetry

$$H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$$

$$\text{real } h[n] \Rightarrow H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}})$$

- Discrete-time frequency response is periodic with period 2π .
- For real coefficients, magnitude is even.
- For real coefficients, phase is odd, except for wrapping and zeros.
- Usually one interval, such as $-\pi \leq \hat{\omega} \leq \pi$, is enough.

Key idea. The whole response is determined by one principal frequency interval.

Delay system

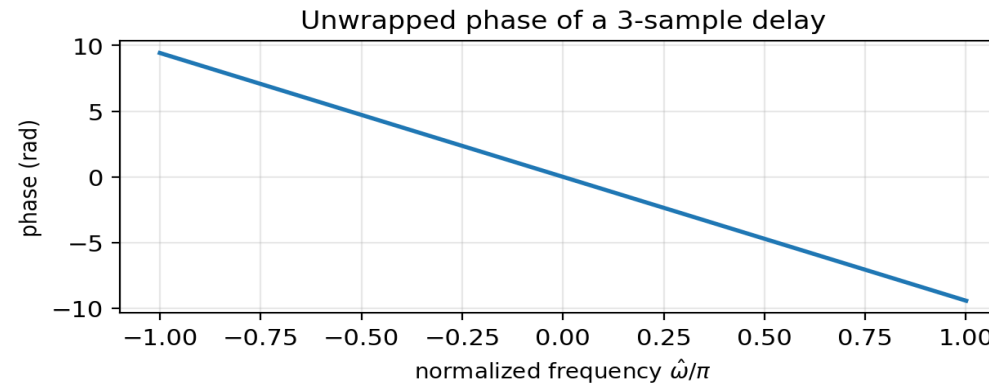
A pure delay changes phase but not magnitude

$$y[n] = x[n - n_0]$$

$$h[n] = \delta[n - n_0]$$

$$H(e^{j\hat{\omega}}) = e^{-j\hat{\omega}n_0}$$

- Magnitude: $|H| = 1$ for all frequencies.
- Phase: $\angle H = -\hat{\omega}n_0$ before wrapping.
- The slope of phase corresponds to delay.



Key idea. A pure delay preserves amplitude while imposing a linear phase shift.

Example 2: pure delay

Compute the output of a delayed sinusoid

$$y[n]=x[n-3], \quad x[n]=2\cos(0.4\pi n+\pi/6)$$

- 1 $H(e^{j\omega}) = e^{-j3\omega}$.
- 2 At $\omega=0.4\pi$: phase shift $= -3(0.4\pi) = -1.2\pi$.
- 3 The amplitude is unchanged: $|H|=1$.
- 4 $y[n]=2\cos(0.4\pi n+\pi/6-1.2\pi)$.

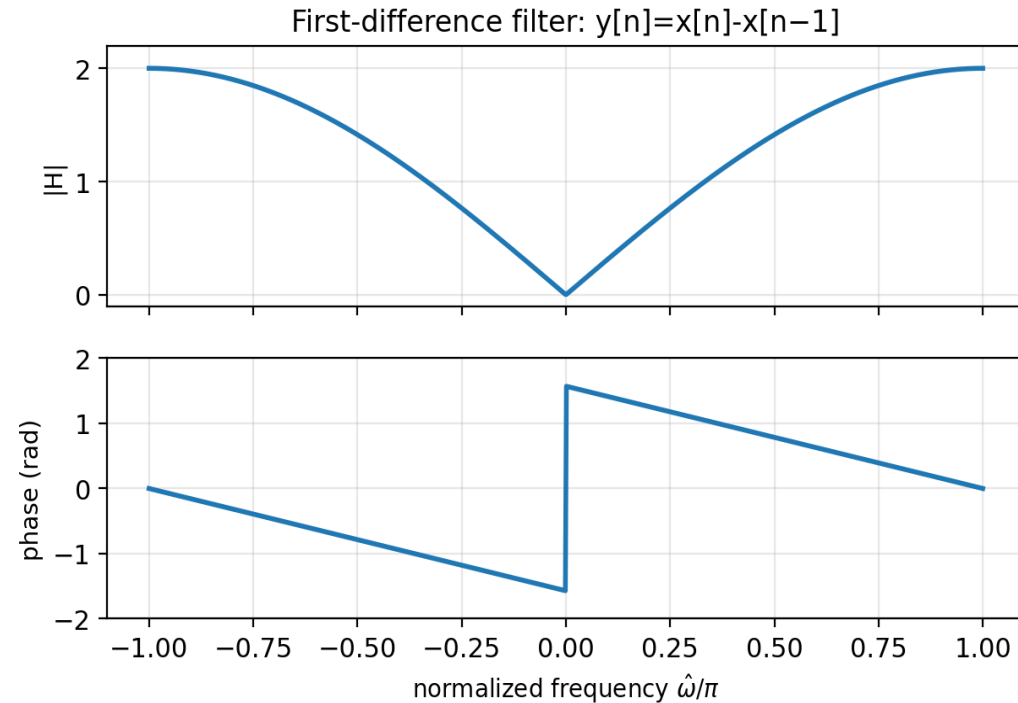
First-difference system

A simple high-pass FIR filter

$$y[n] = x[n] - x[n-1]$$

$$H(e^{j\hat{\omega}}) = 1 - e^{-j\hat{\omega}}$$

$$|H(e^{j\hat{\omega}})| = 2|\sin(\hat{\omega}/2)|$$



Key idea. The first difference removes DC and emphasizes rapid changes.

Example 3: first difference

Evaluate gains and interpret

$$H(e^{j\omega}) = 1 - e^{-j\omega}$$

- 1 At DC: $H(e^{j0}) = 0$, so constant signals are removed.
- 2 At $\omega = \pi/2$: $|H| = 2\sin(\pi/4) = \sqrt{2}$.
- 3 At $\omega = \pi$: $|H| = 2\sin(\pi/2) = 2$.
- 4 Gain increases with frequency $\Rightarrow$ high-pass behaviour.

Cascaded LTI systems

Frequency responses multiply; impulse responses convolve



$$H_{eq}(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}})H_2(e^{j\hat{\omega}})$$

$$h_{eq}[n] = h_1[n] * h_2[n]$$

- Cascade order does not matter for LTI systems.
- Zeros of either subsystem become zeros of the cascade.
- In time domain, the equivalent impulse response is $h_1 * h_2$.

Key idea. Cascades are easier to analyze in the frequency domain because multiplication replaces convolution.

Example 4: cascade

Two moving sums in sequence

$$h_1[n]=\{1,1\}, \quad h_2[n]=\{1,1\}$$

- 1 Frequency responses: $H_1=1+e^{-j\omega}$,
 $H_2=1+e^{-j\omega}$.
- 2 Overall response:
 $H_{eq}=(1+e^{-j\omega})^2$.
- 3 Equivalent impulse response:
 $h_{eq}=h_1*h_2=\{1,2,1\}$.
- 4 The cascade is a three-point triangular smoother.

Key idea. Cascading two simple averagers produces a stronger smoother.

Moving-average filtering

A low-pass smoother with structured zeros

$$y[n] = (1/L) \sum_{k=0}^{L-1} x[n-k]$$

$$h[n] = 1/L, \quad 0 \leq n \leq L-1$$

- A longer average smooths more strongly.
- The main lobe becomes narrower.
- The phase corresponds to a delay of $(L-1)/2$ samples.
- Zeros cancel specific sinusoidal components.

Key idea. A moving average is simple in time domain but very informative in frequency domain.

Moving-average frequency response

Finite geometric series

$$H(e^{j\hat{\omega}}) = (1/L) \sum_{k=0}^{L-1} e^{-j\hat{\omega}k}$$

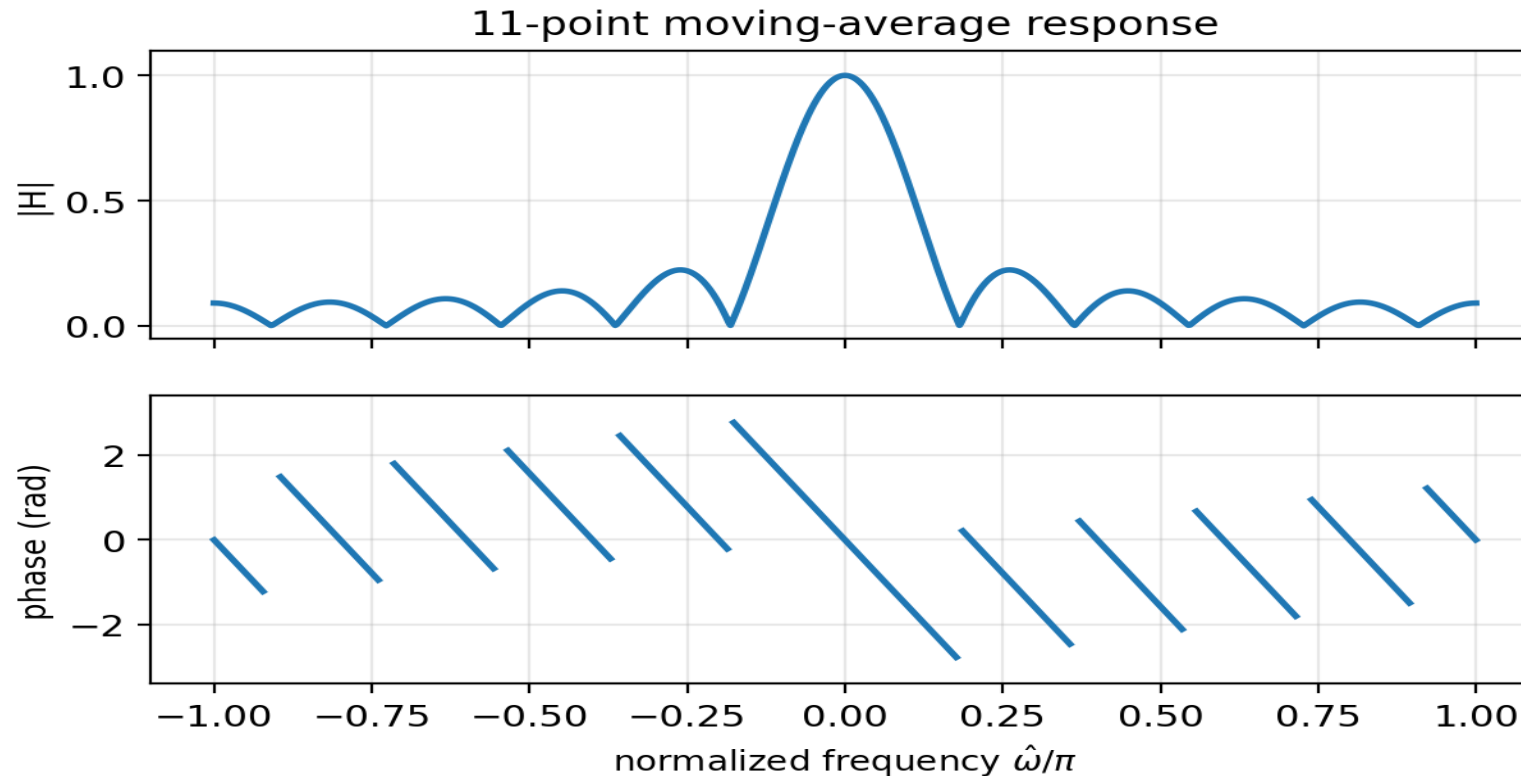
$$H(e^{j\hat{\omega}}) = (1/L) \cdot [\sin(L\hat{\omega}/2) / \sin(\hat{\omega}/2)] \cdot e^{-j\hat{\omega}(L-1)/2}$$

- Amplitude factor: Dirichlet / sinc-like shape.
- Phase factor: linear delay.
- Zeros occur when $\sin(L\hat{\omega}/2)=0$, excluding DC.

Key idea. The closed form separates smoothing strength from delay.

11-point moving-average response

Magnitude response shows low-pass smoothing



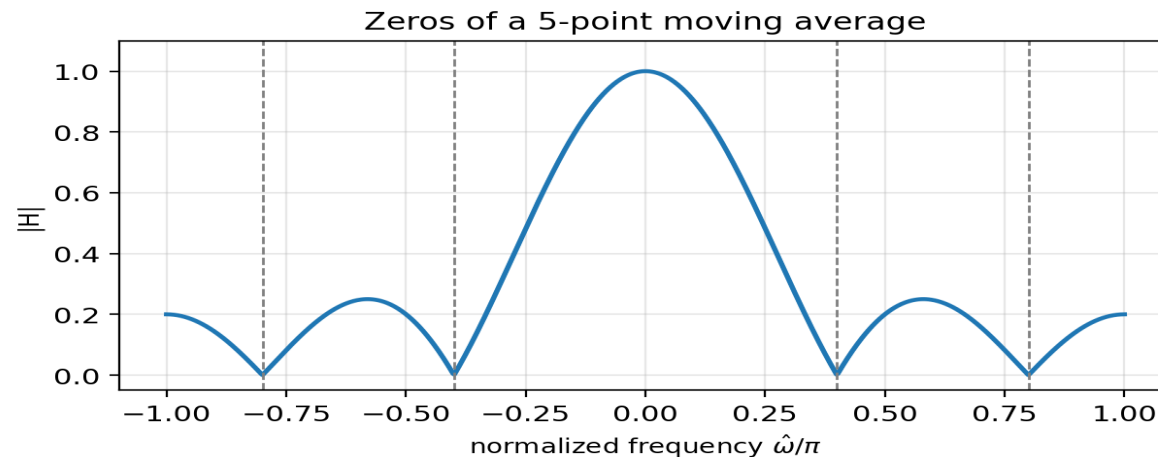
Key idea. Low frequencies near DC pass; many higher frequencies are attenuated or cancelled.

Example 5: moving-average zeros

Cancel a sinusoidal disturbance

Use an L-point moving average to cancel $\hat{\omega} = \pi/2$.

- 1 Zeros satisfy $\hat{\omega} = 2\pi m/L$, with $m=1, \dots, L-1$.
- 2 Set $\pi/2 = 2\pi m/L \Rightarrow L = 4m$.
- 3 Smallest choice: $m=1$, so $L=4$.
- 4 Difference equation: $y[n] = (1/4)(x[n] + x[n-1] + x[n-2] + x[n-3])$.



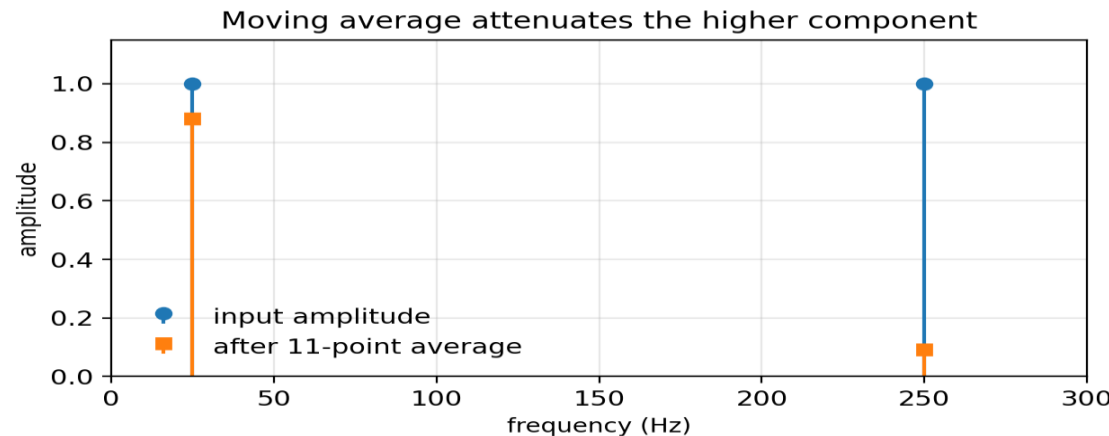
Filtering sampled continuous-time signals

Relating analogue and normalized frequency

$$\hat{\omega} = \omega T_s = 2\pi f / f_s$$

output gain at analogue
frequency f : $|H(e^{j2\pi f/f_s})|$

- Sampling maps analogue frequency into normalized frequency.
- The discrete-time filter acts on that normalized frequency.
- After ideal reconstruction, the analogue frequency is preserved if there is no aliasing.



Key idea. Discrete-time frequency response predicts analogue behaviour after sampling and reconstruction.

Example 6: sampled signal

11-point average at $f_s = 1000$ Hz

$$x(t) = \cos(2\pi \cdot 25t) + \cos(2\pi \cdot 250t)$$

- 1 Normalized frequencies: $\hat{\omega}_1 = 2\pi \cdot 25 / 1000 = 0.05\pi$; $\hat{\omega}_2 = 0.5\pi$.
- 2 Evaluate the 11-point moving-average response at both frequencies.
- 3 The 25-Hz component lies near DC and is mostly preserved.
- 4 The 250-Hz component lies in a low-gain region and is strongly attenuated.

Even- and odd-length moving averages

Delay can be integer or fractional

linear phase factor: $e^{-j\omega(L-1)/2}$

delay = $(L-1)/2$ samples

- Odd L gives an integer-sample delay, e.g., $L=5 \rightarrow$ delay 2.
- Even L gives a half-sample delay, e.g., $L=4 \rightarrow$ delay 1.5.
- A half-sample delay is meaningful after continuous-time reconstruction.

Key idea. Linear phase preserves waveform shape apart from gain and delay.

Notation and takeaways

Minimum language for frequency response

Symbol	Meaning
$x[n], y[n]$	Input and output discrete-time sequences
$h[n]$	Impulse response of the FIR filter
b_k	FIR filter coefficient at delay k
M	Filter order; the largest delay index
$L=M+1$	Filter length or number of coefficients
$H(e^{j\hat{\omega}})$	Frequency response evaluated at normalized radian frequency $\hat{\omega}$
$ H(e^{j\hat{\omega}}) $	Magnitude response or frequency-dependent gain
$\angle H(e^{j\hat{\omega}})$	Phase response
$\hat{\omega}$	Normalized radian frequency, in radians per sample
ω	Continuous-time radian frequency, in radians per second
T_s	Sampling period
$f_s=1/T_s$	Sampling rate
$*$	Convolution operator
$\delta[n]$	Unit impulse sequence
$u[n]$	Unit-step sequence